DO NOW — RECALL

Thinking logically: decisions and program flow

Name: _____ Class: _____ Date: _____

1. Why must a booking be saved before confirmation?

2. What is true AND false?

3. What does an if condition control?

4. What is false OR true?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Confirmation must reflect a successful stored change.

2. Model answer

False.

3. Model answer

Which branch of the program runs.

4. Model answer

True.

Vocabulary — know these terms

condition — Expression evaluated as true or false.

branch — Path selected by a decision.

AND — True only if both operands are true.

OR — True if either or both operands are true.

truth table — All Boolean input combinations and outputs.

NEW LEARNING

Read, annotate and explain

Locate the decision

A discount needs BOTH an eligible membership and a basket of at least £50.

Follow the true branch

For member=true and basket=60, the condition is true and 10% discount produces £54.

Follow the false branch

For member=false with basket=60, the AND expression is false, so the full £60 is charged.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A cinema permits admission for age≥15 AND a valid ticket. Trace age=16 with no ticket.

Predict first

The age condition is true, but the ticket condition is false.

Model the reasoning

AND requires both true; the result is false, so the deny branch executes.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give A AND B for (false,false), (false,true), (true,false), (true,true).

Question 2

Access is allowed if Year13 AND (HasCode OR TeacherApproved). Evaluate Year13=true, HasCode=false, TeacherApproved=true; then Year13=false, HasCode=true, TeacherApproved=true.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A developer changes the discount condition `member && basket>=50` to `member || basket>=50`. Give two wrong outcomes and correct the expression.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Replacing AND with OR in a two-part eligibility rule can permit extra cases.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.